Task 3: Reflection on AI Paradoxes

Through this assignment, I gained a deeper understanding of the paradoxes of artificial intelligence, especially the relationship between AI capabilities, reliability, security, and human control. By developing a CV verification agent and then attempting to deceive it, I experienced how the same AI technology can be both useful and vulnerable.

1. The Paradox of AI Efficiency and Reliability

The first paradox is that AI can significantly improve efficiency without necessarily guaranteeing accuracy. In Task 1, I developed an AI agent using LangChain and DeepSeek to verify CV information against LinkedIn and Facebook profiles through an MCP server. Compared with manual verification, the agent can automatically retrieve information, compare candidate backgrounds, and generate reliability scores.

However, my agent correctly classified only 5 out of 7 CVs in the public test, achieving an accuracy of 71.43%. It incorrectly rejected one valid CV and failed to identify a false skill claim in another. These mistakes demonstrate that although AI can automate complex tasks, its judgments are not always reliable.

This made me realize that AI automation does not eliminate errors. Instead, it may introduce new types of errors that are difficult for users to notice, especially when the system presents its results confidently.

2. The Paradox of AI Security and Vulnerability

The second paradox is that AI can be used to improve security while simultaneously creating new security risks.

In Task 1, my AI agent acted as a verifier, attempting to detect false information in candidates' CVs. However, in Task 2, I became the attacker and tried to manipulate the same verification process.

I created an adversarial CV for Kelly Tsang by adding Python as a skill that was not listed in her LinkedIn profile. I also included a fabricated automated verification record containing instructions intended to influence the agent's judgment.

Surprisingly, my verification agent assigned this adversarial CV a reliability score of 0.95, even though it contained a false claim according to the assignment's verification criteria.

However, this experiment does not conclusively prove that the prompt injection caused the failure. The agent might also have overlooked the additional skill because its verification process was insufficiently thorough.

Nevertheless, the experiment revealed an important security issue: AI agents must process external information to complete their tasks, but that information may contain malicious instructions. If an agent cannot distinguish trusted instructions from untrusted document content, its decisions may be manipulated.

3. The Paradox of AI Autonomy and Human Control

The third paradox concerns the relationship between AI autonomy and human control.

AI agents are powerful because they can independently select tools, retrieve information, and make decisions. In this assignment, my agent could automatically communicate with an MCP server and analyze candidate information without requiring manual intervention at every step.

However, greater autonomy also introduces uncertainty. The agent may select an incorrect candidate, overlook important evidence, or misunderstand the information returned by external tools. These errors can affect the final judgment without being immediately visible to the user.

This is particularly important in real-world applications such as recruitment, banking, and Know Your Customer (KYC) verification. An incorrect judgment could unfairly reject a qualified candidate or allow misleading information to pass verification.

Therefore, I believe AI agents should not be given unlimited decision-making authority in high-stakes situations. Human supervision, transparent reasoning, and reliable verification mechanisms remain essential.

4. The Paradox of AI Confidence and Actual Correctness

Another interesting paradox I observed is that an AI system's confidence does not necessarily reflect the correctness of its decision.

In Task 2, the adversarial CV received a score of 0.95. According to my agent's scoring rules, this value represented a highly reliable CV. However, the CV intentionally contained information that contradicted the verification data.

This shows that a high numerical confidence score does not automatically mean that the underlying conclusion is correct. In my implementation, the score was generated by the language model according to predefined scoring instructions rather than calculated from a statistically calibrated probability model.

Therefore, users should be cautious when interpreting AI-generated confidence scores. A reliable system should provide supporting evidence and explain which claims have been verified, rather than relying only on a single numerical score.

5. Conclusion and Personal Reflection

Overall, this assignment changed my understanding of AI agents. Before completing these tasks, I mainly focused on how to make an AI agent successfully execute its intended functions. However, after implementing a verification agent and testing an adversarial attack, I realized that building an agent that works is different from building an agent that can be trusted.

In my opinion, improving AI reliability requires several important measures, including stronger identity verification, claim-by-claim evidence checking, better protection against prompt injection, and more transparent decision-making processes.

Most importantly, AI should be treated as a tool that supports human judgment rather than an authority that is automatically correct.

This assignment helped me understand that the future development of AI agents should focus not only on improving their capabilities and efficiency, but also on ensuring their security, reliability, and accountability.